AntClust

A clustering algorithm based on the chemical recognition system of ants

Netlogo Implementation and Analysis

Presentation Outline

- **Biological foundations** behind the paper
- **The AntClust paper** by Labroche et al.
- **Implementation in NetLogo**.
- **Datasets used**
- **Representative runs**: to understand the «step by step» behaviour
- **Aggregated results**: to understand the impact of parameters
- **Conclusions**

Biological Foundations

Real ants solve a daily recognition problem to protect the nest. They form groups sharing a common chemical identity to distinguish nestmates from intruders.

Identification signals are stored on the body. This identity is dynamic and constantly updated through social contact.

There is no central control; each ant maintains its own individual "vision" of the colony odor.

AntClust Paper

The algorithm translates biological behavior into 5 computational phases:

- **Initialization:** Creating agents and assigning data.
- **Learning (Ontogenesis):** Setting initial acceptance thresholds.
- **Interactions:** Running the social rules (R1–R6).
- **Nest Cleanup:** Removing small, unstable clusters (noise).
- **Finalization:** Assigning remaining "orphan" agents.

These local stochastic interactions allows a global clustering to emerge

Data is clustered without knowing the expected number of clusters in advance

AntClust Paper: Initialization

Each data object is associated with one artificial ant; the data vector (no limitations on data types) represents the ant's genetic information.

Ants can calculate their similarity versus another ant as an average of the similarity for each data type.

$$Sim_{type_i}(a_i, a_j) = 1 - \left(\frac{1}{occ(type_i)} \times \sum_{k=1}^{occ(type_i)} \Delta_{type_i}(a_i, a_j) \right)$$

AntClust Paper: Initialization

Confidence estimators M and M^+ are set to 0.

M: ant's individual perception of its nest's size and social importance.

M^+: ant's individual perception of the degree of acceptance and integration within its current nest.

AntClust Paper: Learning

This phase emulates how young ants that learn their colony odor accept or reject others.

Each ant learns an Acceptance Threshold (T_i) during a set number of random meetings.

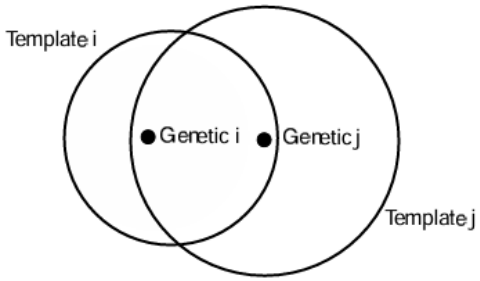
$$T_i = \frac{avg\big(Sim(i,\cdot)\big) + \max\big(Sim(i,\cdot)\big)}{2}$$

AntClust Paper: Interactions

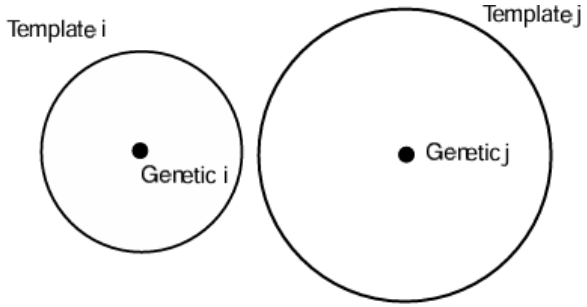
For a certain number of iterations, two random ants are selected to interact with each other and build nests.

Their template is updated and they evaluate each other to determine if they accept each other or not.

$$Acceptance(i,j) \Leftrightarrow (Sim(i,j) > Template_i)\ and\ \left(Sim(i,j) > Template_j\right)$$



Acceptance



Rejection

Interactions: Constructive Rules

All variables updates are handled with the formula:

$increase: x \leftarrow (1-\alpha) \cdot x + \alpha$ $\qquad$ $decrease: x \leftarrow (1-\alpha) \cdot x$

R1 (Creation)

If both ants accept each other and are unassigned

They create a new nest

R2 (Assimilation)

If both ants accept each other and one is unassigned

The unassigned ant joins the nest

R3 (Reinforcement)

If both ants accept each other and are from the same cluster

Their confidence increases (M, M^{+})

Interactions: Corrective Rules

All variables updates are handled with the formula:

$increase: x \leftarrow (1-\alpha) \cdot x + \alpha$ $\qquad$ $decrease: x \leftarrow (1-\alpha) \cdot x$

R4 (Conflict)	R5 (Competition)	R6 (Default)
If both ants reject each other and are from the same nest.	If both ants accept each other and are from different nests.	If no other rule applies
Their confidence decreases (M^+) and the one with lowest M^+ loses its nest	Their confidence decreases (M) and the one with lowest M joins the other ant's nest.	No action is taken

AntClust Paper: Cleanup

This phase filters out statistical artifacts and noise generated during the interaction phase.

Nests are evaluated based on their total population size and must exceed a minimum size threshold to be preserved.

Members of suppressed nests lose their identity (Label = 0) and return to the "orphan" pool for final reclassification.

AntClust Paper: Finalization

This phase forces orphans to be assigned to a nest.

Each unassigned ant performs 1NN (finds its most similar neighbour) among those that already have a valid label, and it takes its label.

NetLogo Implementation: Architecture

Agents do not move; each turtle represents a fixed data point

Coordinates are determined by the first two attributes of the dataset.

While mapped in 2D for visual feedback, similarity is computed across the full attribute set.

Clustering convergence is observed through color changes rather than movement.

NetLogo Implementation: Heuristics

Several decisions have been taken to transform the theoretical algorithm into a real model:

- The original paper does not specify the number of initial meetings. It has been implemented as a percentage of the total population ($\mathrm{initial} - \mathrm{meetings} - \mathrm{pct} \times \mathrm{N}$) to ensure the template scales with the dataset size.
- The original paper does not specify the minimum population needed for a cluster to survive. This has been implemented as a slider to discover different behaviours by varying it.

NetLogo Implementation: Enhancements

The following enhancements were introduced to optimize clustering stability and explore the algorithm behaviour:

- **Learning Rate** (α): this slider controls how quickly the social estimators are updated during meetings. The original paper suggests a fixed value of 0.2, this implementation allows for tuning of the system's learning speed.
- **Memory Decay** (λ): a custom addition that reduces the maximum similarity value over time. This prevents the acceptance threshold from "freezing" at a very high level if an agent encounters an outlier early in the simulation.
- **Number of Iterations**: the original paper used a fixed 300,000 iterations, this implementation calculates them dynamically ($\mathrm{N} \times \mathrm{iterations} - \mathrm{multiplier}$) with a slider. This follows updated literature suggesting $75 \times \mathrm{N}$ as a more efficient standard.

NetLogo Implementation: Plots

Threshold plot:

Visualizes the distribution of acceptance criteria among the agents

Cluster plot:

Monitors the number of active nests (clusters) over time.

Demonstrates how the agents merge or split groups

NetLogo Implementation: Metrics

NetLogo is connected to the scikit-learn library via the py extension. This allows for the calculation of complex statistics that NetLogo cannot handle alone.

Adjusted Rand Index (ARI): Measures the similarity between the found clusters and the real labels while correcting for groupings that could happen by random chance.

Silhouette: Measures how well-separated the clusters are by comparing the distance of a point within its own group to its distance from other groups

Homogeneity: Evaluates if each found cluster contains only data points that belong to a single real class. (precision)

Completeness: Evaluates if all data points belonging to the same real class are successfully assigned to the same cluster. (recall)

Datasets: Iris

Characteristics

objects: 150 classes: 3 dimensions: 4

Chosen to verify if the algorithm can separate well-known, simple clusters with a low number of attributes.

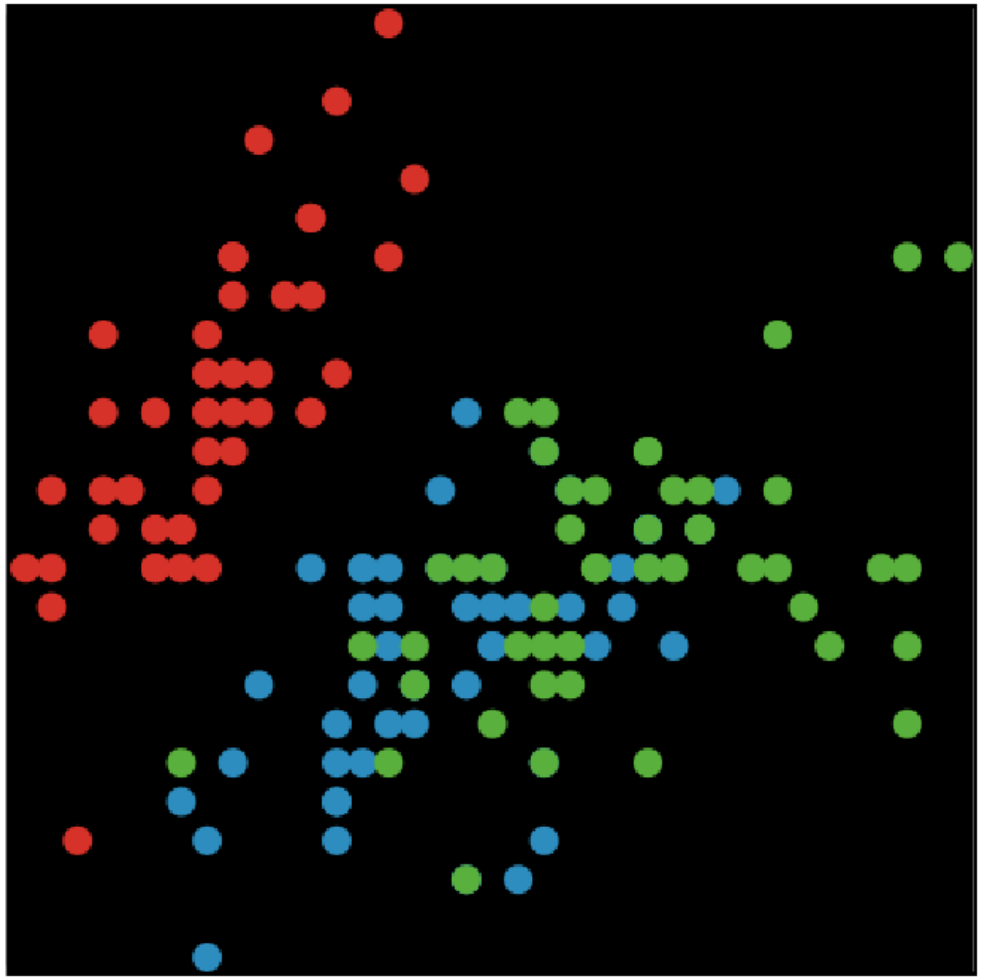
Ground Truth:

Silhouette: 0.582

KMeans Comparison:

ARI: 0.716 | homogeneity: 0.736 |

completeness: 0.747 | silhouette: 0.504



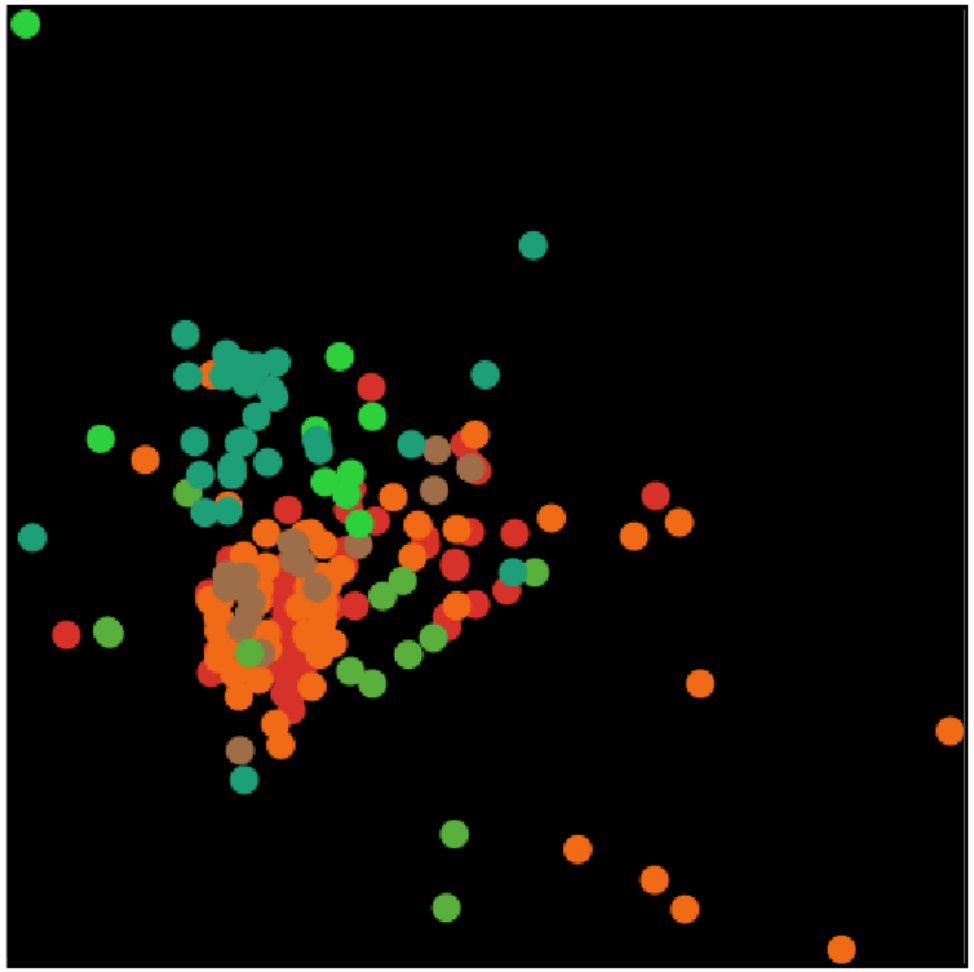
setosa, **versicolor**, **virginica**

Datasets: Glass

Characteristics

objects: 214 classes: 6 dimensions: 9

Chosen because it is a difficult dataset with many classes and overlapping values, used to test the algorithm when similar groups are present.



Ground Truth:

Silhouette: - 0.091

KMeans Comparison:

ARI: 0.167 | homogeneity: 0.320 |

completeness: 0.332 | silhouette: 0.380

Datasets: Breast Cancer

Characteristics

objects: 569 classes: 2 dimensions: 30

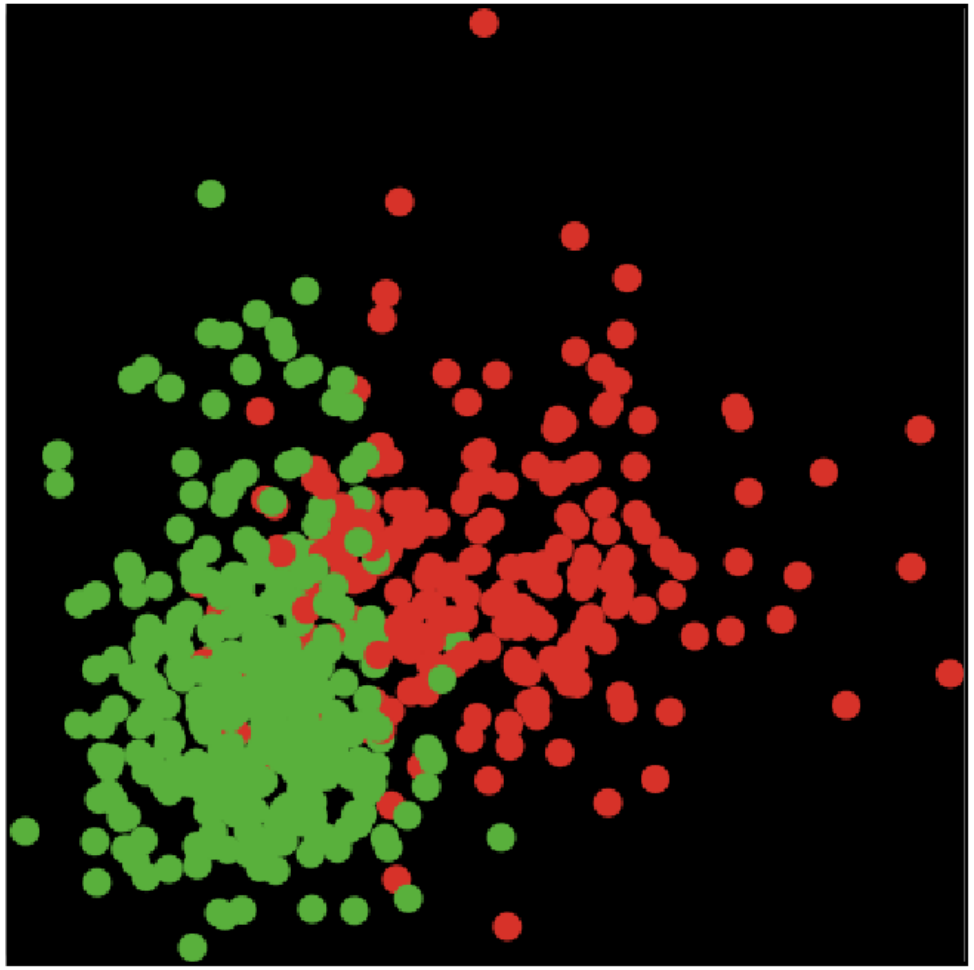
Chosen to test the algorithm in a high-dimensional space (curse of dimensionality)

Ground Truth:

Silhouette: 0.514

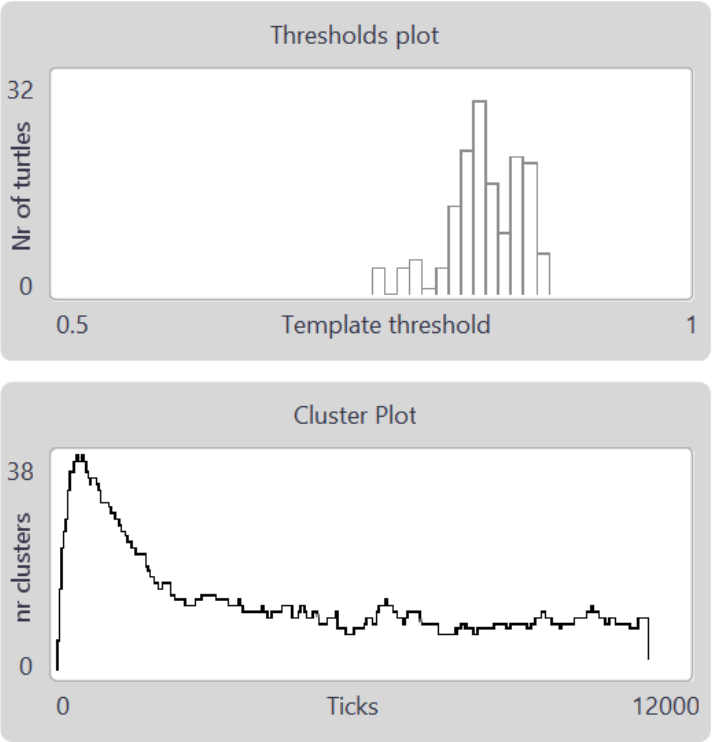
KMeans Comparison:

ARI: 0.730 | homogeneity: 0.611 | completeness: 0.635 | silhouette: 0.385

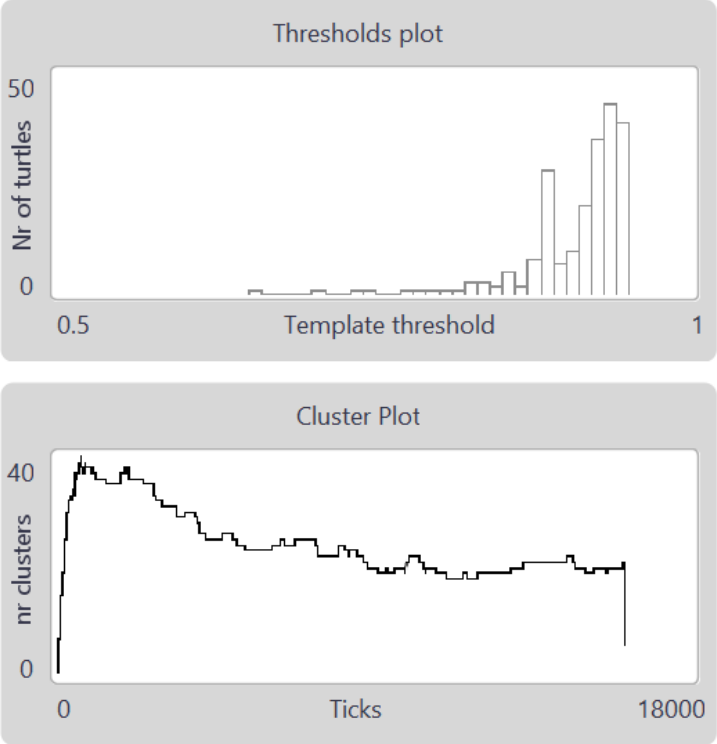


benign, malign

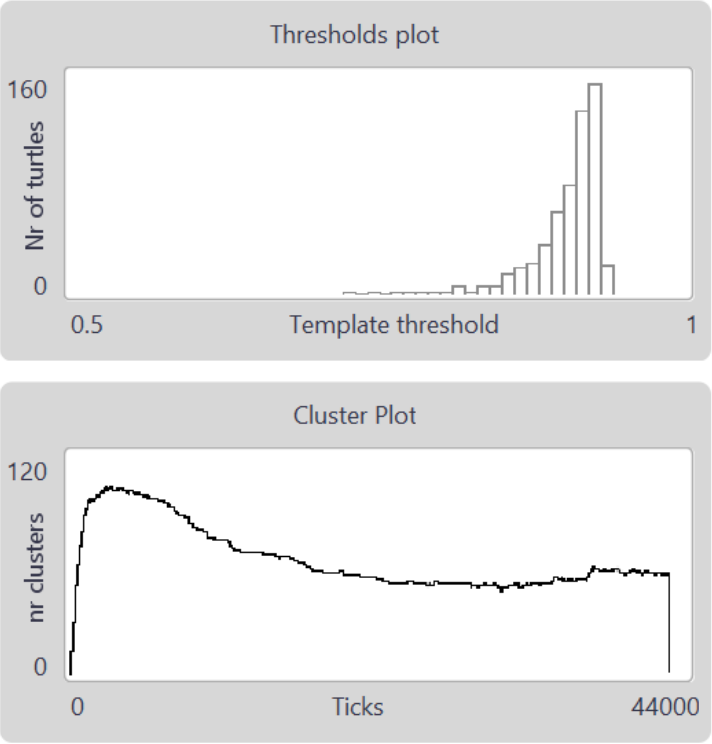
Experiments: Representative Runs



Iris



Glass



Breast-Cancer

Plots after the cleanup phase

Experiments: Aggregated Results

The experimental phase focused on parameter calibration and the testing of the model.

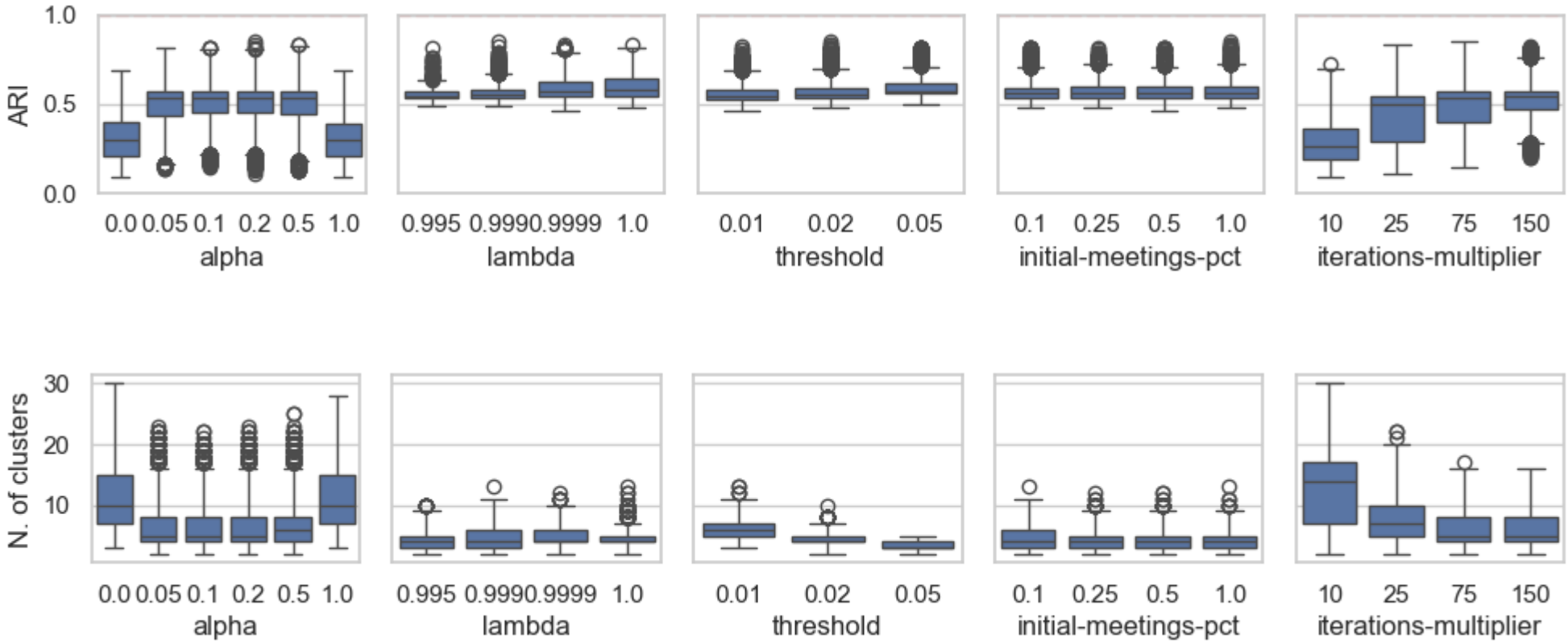
Grid Search performed using NetLogo's BehaviorSpace to test thousands of combinations of parameters.

The algorithm is stochastic, results from a single run are insufficient. Data were aggregated over 20 **independent runs** for each configuration.

Results from each dataset used to fine-tune following analysis

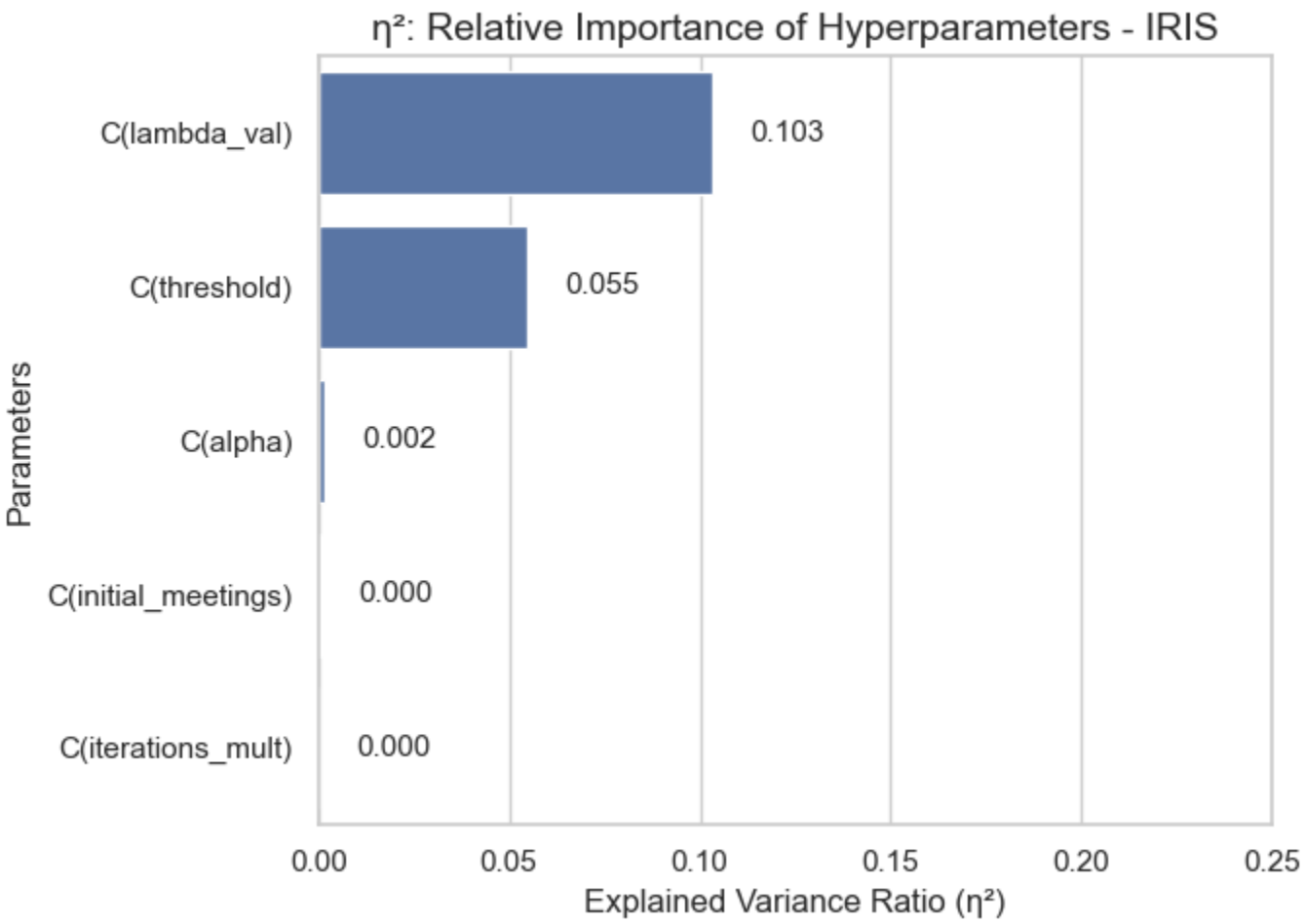
- Executed around 30K simulations
- Collected: ARI | homogeneity | completeness | silhouette | number of clusters.

Aggregated Results: Iris



Some form of memory is essential

Aggregated Results: Iris



Alpha = 0, alpha = 1 excluded, Iterations-multiplier >= 75.

Lambda is the driver of the algorithm.

Still, its impact is small. For easy datasets the hyperparameters are not that important

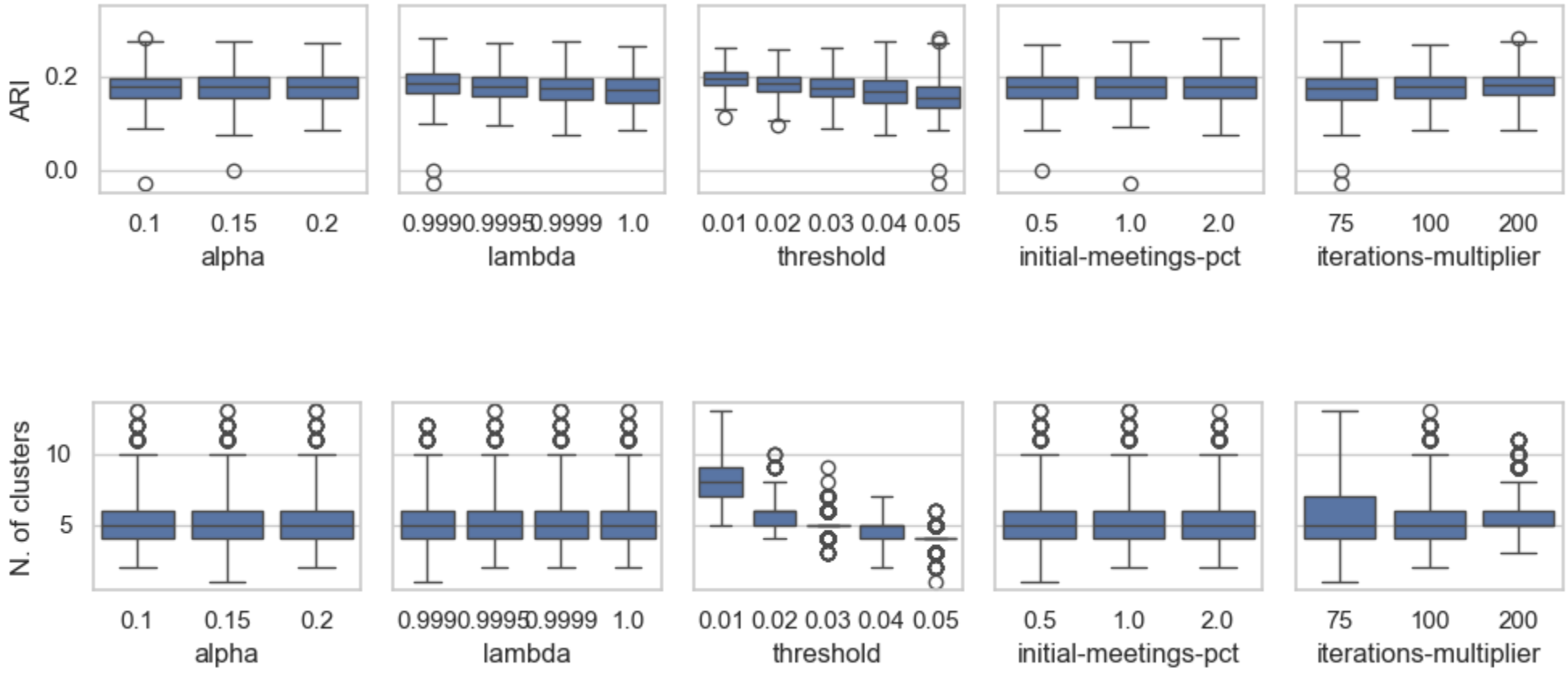
Aggregated Results: Iris

lambda	threshold	res-ari	res-p	res-r	res-s	n_clusters
1.000	0.05	0.618	0.702	0.695	0.440	3.441
1.000	0.05	0.611	0.695	0.698	0.439	3.412
1.000	0.02	0.596	0.720	0.628	0.356	4.581
1.000	0.02	0.588	0.714	0.630	0.354	4.548
0.999	0.05	0.585	0.659	0.752	0.466	3.134

KMeans Comparison: ARI: 0.716 | homogeneity: 0.736 | completeness: 0.747 | silhouette: 0.504

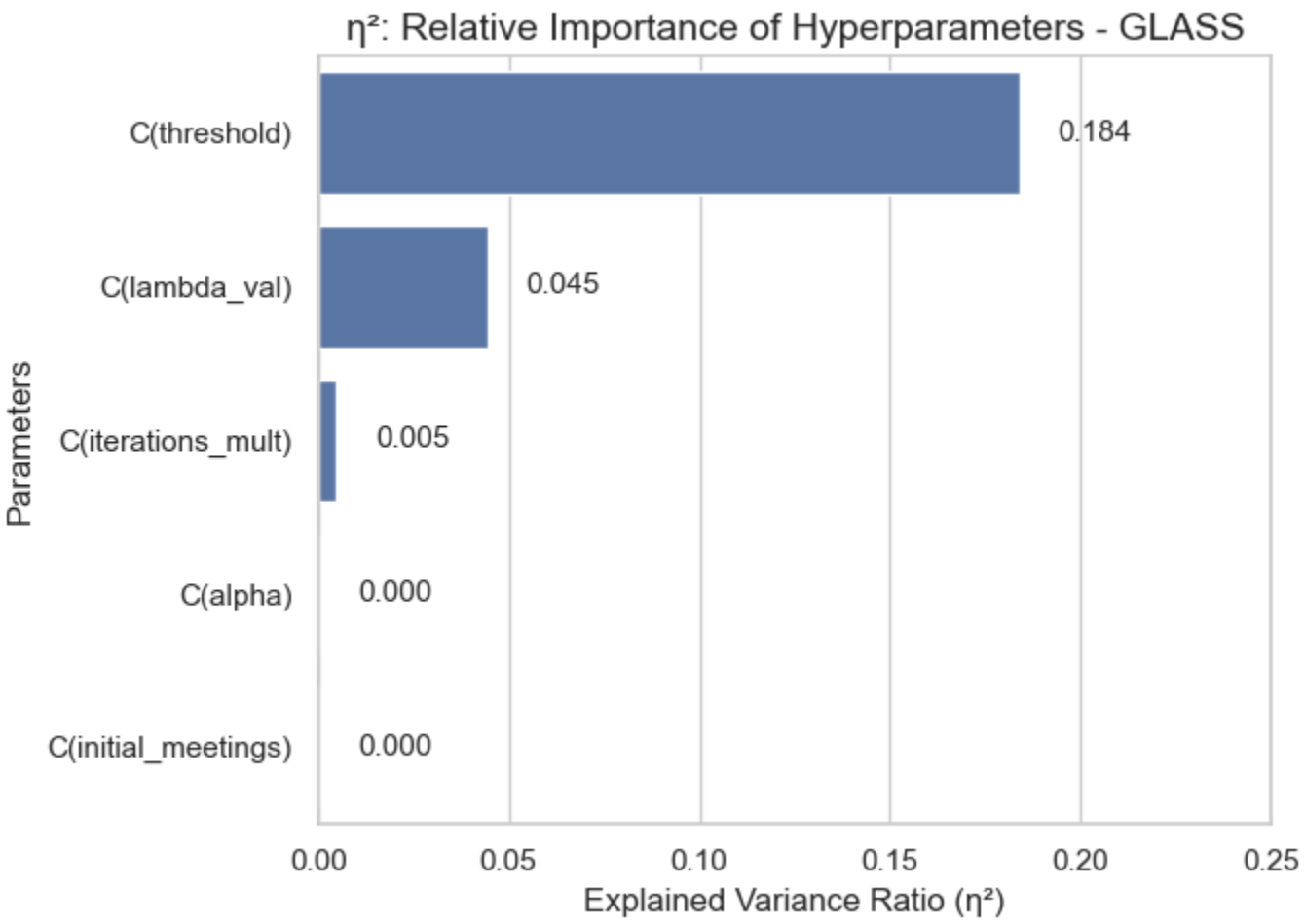
The algorithm captures the structure of data, but prefers to separate data rather than mixing it

Aggregated Results: Glass



Lambda and Threshold are the most important parameters. Low performance across the line

Aggregated Results: Glass



GLASS is a very imbalanced dataset

Class 6 has 9 elements

A low threshold preserves all clusters

The algorithm still manages to give a structure to the data regardless of the threshold

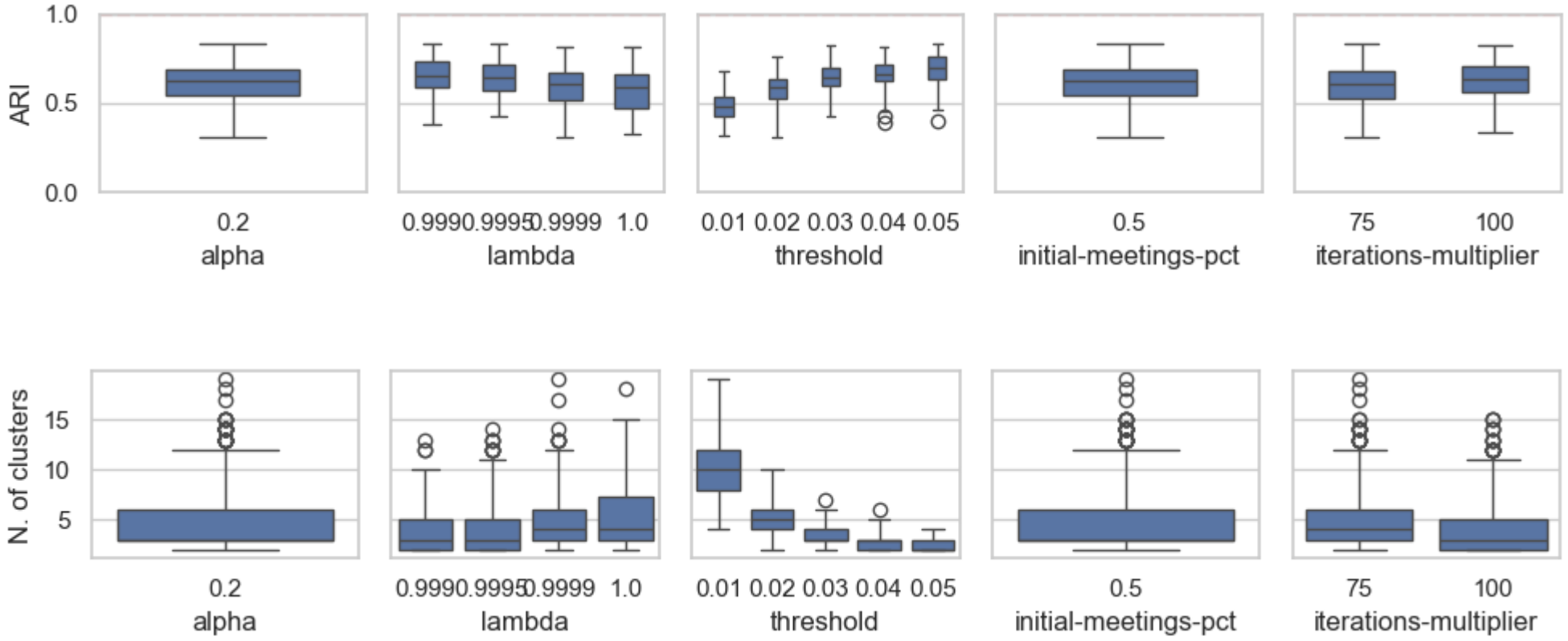
Aggregated Results: Glass

lambda	threshold	res-ari	res-p	res-r	res-s	n_clusters
0.999	0.01	0.198	0.350	0.345	0.079	8.060
1.000	0.01	0.196	0.354	0.344	0.080	7.991
1.000	0.01	0.193	0.372	0.345	0.078	8.478
1.000	0.01	0.192	0.362	0.343	0.080	8.161
0.999	0.02	0.187	0.313	0.338	0.102	5.961

KMeans Comparison: ARI: 0.167 | homogeneity: 0.320 | completeness: 0.332 | silhouette: 0.380

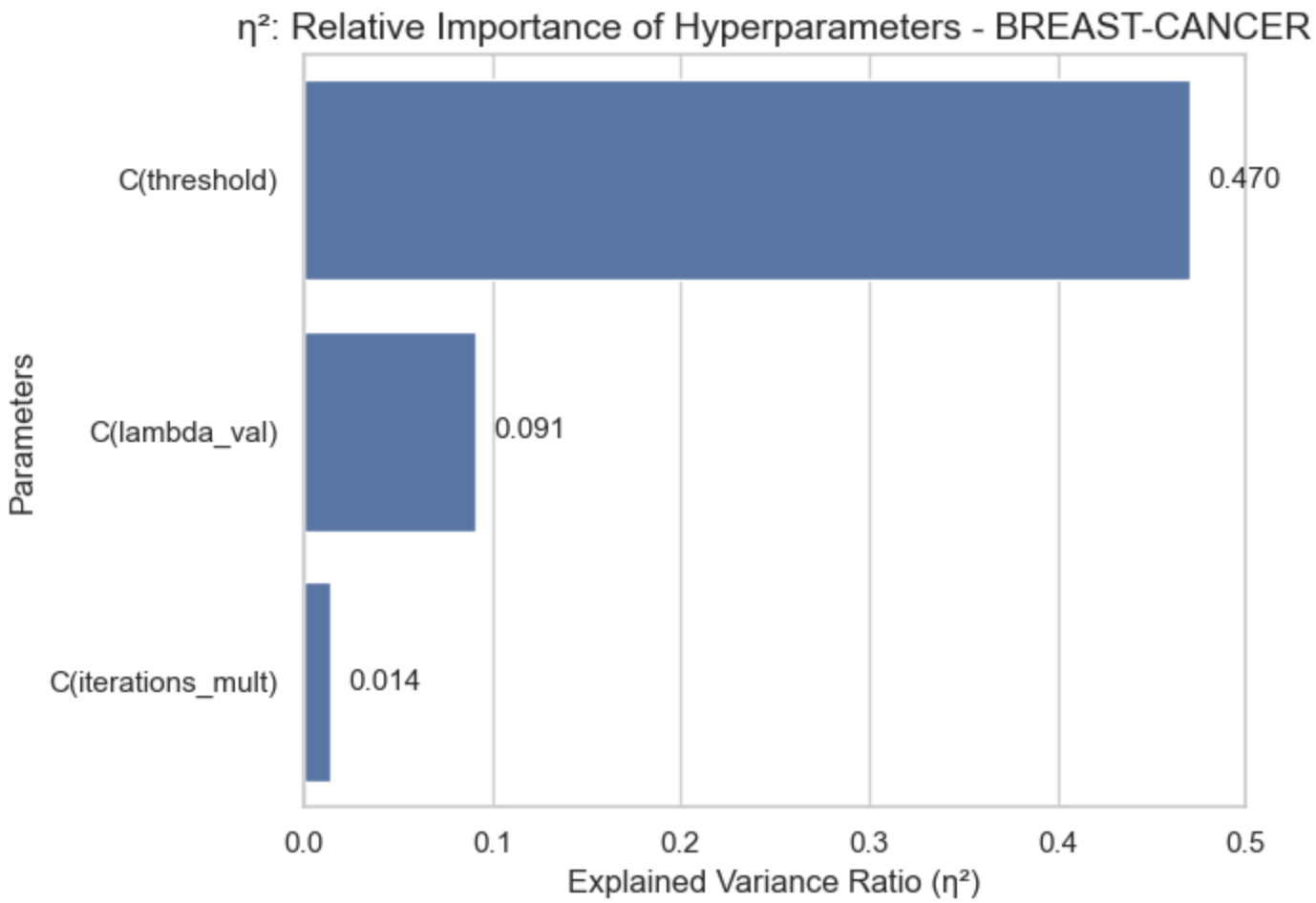
The model prefers to create new clusters instead of mixing data points

Aggregated Results: Breast Cancer



Efficient scaling in high dimensional data

Aggregated Results: Breast Cancer



Threshold is the driver of the convergence

Lambda stabilizes the results

Aggregated Results: Breast Cancer

lambda	threshold	res-ari	res-p	res-r	res-s	n_clusters
0.999	0.05	0.725	0.635	0.612	0.489	2.275
1.000	0.05	0.700	0.627	0.571	0.453	2.500
1.000	0.05	0.696	0.619	0.588	0.478	2.350
0.999	0.03	0.695	0.633	0.541	0.390	2.850
1.000	0.04	0.687	0.621	0.538	0.424	2.650

KMeans Comparison: ARI: 0.730 | homogeneity: 0.611 | completeness: 0.635 | silhouette: 0.385

The system accurately identifies the 2 real classes with a high ARI

Conclusions

- The algorithm finds the natural data structure without needing a predefined number of clusters, as accurately as standard Kmeans
- Social interaction rules mimic ant behavior to group similar objects without central control.
- Good performance and stability are maintained even in high-dimensional spaces.
- Threshold and memory decay are the primary drivers of cluster convergence.